The first 5 questions
2.5 points for correct answer
2.5 points for correct explanation

CS556-A Fall 2023 Midterm

Name:

Closed book: no textbook, no electronic devices, one sheet of paper with handwritten notes.

Read carefully before answering! Write your answers on the white sheets given to you. Show all your work, not only the final answer.

Question 1 *(5 points)* Can a system of linear equations have exactly three solutions? Explain.

No. A system can have:
- No solution
- One unique solution
- Infinite number of solutions (line, place etc)

Question 2 *(5 points)* Let A and B be two arbitrary $2x2$ matrices. If a vector x is in the null space of A, can we say also that x is in the null space of BA? Explain.

$Ax = 0$

$(BA)x = B(Ax) = B(0) = 0 \rightarrow x \in N(BA)$

YES.

Question 3 *(5 points)* Is it possible for the nullspace of a matrix not to include the zero vector? Explain.

No, it is not possible. Null space always passes through the origin.

$Ax = 0$ when $x = 0$.

Question 4 *(5 points)* If a system of linear equations has two different solutions, it must have infinitely many solutions. True or False? Explain.

True. A system has i) no solution, ii) unique solution or iii) infinite number of solutions.
So, if it has 2, then it has an infinite number of solutions

Question 5 *(5 points)* Any system of n linear equations in a n variables has at most n solutions. True or False?

False. A system with n variables can have an infinite number of solutions.

Question 6 *(25 points)* Let $x_1 = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}$, $x_2 = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$, and $v = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$.

a) Find a vector x_3 such that $\{x_1, x_2, x_3\}$ is a linearly independent set of vectors.

b) Find the determinant of the 3-by-3 matrix composed of columns x_1, x_2, x_3.

c) Find the rank of this matrix.

d) Find the inverse of this matrix.

e) Find a condition that a, b, c must satisfy for the set of the vectors $\{x_1, x_2, v\}$ to be linearly dependent.

6 points
1pts for correct method, +2.5 for det
3- Correct rank, 0.5 for reasoning
4- correct approach, +2 correct inverse
4 correct approach +2 for correct answer.

a) In order for the set $\{x_1, x_2, x_3\}$ to be linearly independent, then matrix $A = [x_1\ x_2\ x_3]$ should have a determinant different from 0.

Let $x_3 = \begin{bmatrix} 1 \\ -1 \\ 6 \end{bmatrix} \Rightarrow A = \begin{bmatrix} 2 & -1 & 1 \\ -3 & 1 & -1 \\ 4 & 2 & 6 \end{bmatrix}$

$$\det(A) = 1\cdot\begin{vmatrix} -3 & 1 \\ 4 & 2 \end{vmatrix} - (-1)\begin{vmatrix} 2 & -1 \\ 4 & 2 \end{vmatrix} + 6\begin{vmatrix} 2 & -1 \\ -3 & 1 \end{vmatrix}$$

$$= -10 + 8 - 6 = -8 \neq 0$$

Hence, the set of vectors is linearly independent

b) $\det(A) = -8$ given from the previous solution on a.

c) $\text{rank}(A) = 3$ since all three vectors are linearly independent

d) Using Gauss-Jordan

$$\left[\begin{array}{ccc|ccc} 2 & -1 & 1 & 1 & 0 & 0 \\ -3 & 1 & -1 & 0 & 1 & 0 \\ 4 & 2 & 6 & 0 & 0 & 1 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 0 & -7/4 & -1 & 1/8 \\ 0 & 0 & 1 & 5/4 & 1 & 1/8 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} -1 & -1 & 0 \\ -7/4 & -1 & 1/8 \\ 5/4 & 1 & 1/8 \end{bmatrix}$$

e) In order for the set to be linearly dependent $\det(A) = 0$

$$A = \begin{bmatrix} 2 & -1 & a \\ -3 & 1 & b \\ 4 & 2 & c \end{bmatrix} \Rightarrow \det(A) = a\begin{vmatrix} -3 & 1 \\ 4 & 2 \end{vmatrix} - b\begin{vmatrix} 2 & -1 \\ 4 & 2 \end{vmatrix} + c\begin{vmatrix} 2 & -1 \\ -3 & 1 \end{vmatrix}$$

$$= -10a - 8b - c = 0$$

$$\boxed{10a + 8b + c = 0}$$

Question 7 *(25 points)* Given matrix $A = \begin{bmatrix} 1 & 0 & 1 & 0 & 2 \\ 2 & 1 & 1 & 1 & 3 \\ 1 & 3 & -1 & 2 & 2 \\ 0 & 3 & -2 & 2 & 0 \end{bmatrix}$.

a) Find $N(A)$ (the nullspace of A). 5-Row Reduction, 2 for each special sol, 1 x complete

b) What is the dimension of $N(A)$? 4 pts

c) Determine 3 vectors that are in the nullspace of A. 2pts for each vector

d) Calculate the inner product of these 3 vectors with one of the rows of A. 5pts (Reasoning or calc)

a) $\begin{pmatrix} 1 & 0 & 1 & 0 & 2 \\ 2 & 1 & 1 & 1 & 3 \\ 1 & 3 & -1 & 2 & 2 \\ 0 & 3 & -2 & 2 & 0 \end{pmatrix} \xrightarrow[R_3 = R_3 - R_1]{R_2 = R_2 - 2R_1} \begin{pmatrix} 1 & 0 & 1 & 0 & 2 \\ 0 & 1 & -1 & 1 & -1 \\ 0 & 3 & -2 & 2 & 0 \\ 0 & 3 & -2 & 2 & 0 \end{pmatrix} \xrightarrow[R_4 = R_4 - 3R_2]{R_3 = R_3 - 3R_2}$

$\begin{pmatrix} 1 & 0 & 1 & 0 & 2 \\ 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -1 & 3 \end{pmatrix} \xrightarrow{R_4 = R_4 - R_3} \begin{pmatrix} \textcircled{1} & 0 & 1 & 0 & 2 \\ 0 & \textcircled{1} & -1 & 1 & -1 \\ 0 & 0 & \textcircled{1} & -1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ x & y & z & w & r \end{pmatrix}$

2 free variables w and r.

Special solution 1 (S1), we set $w = 1$ and $r = 0 \rightarrow z = 1, y = 0, x = -1$

$$S1 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

Special solution 2 (S2), we set $w = 0$ and $r = 1, \rightarrow z = -3, y = -2, x = 1$

$$S2 = \begin{bmatrix} 1 \\ -2 \\ -3 \\ 0 \\ 1 \end{bmatrix}$$

$$N(A) = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} t + \begin{pmatrix} 1 \\ -2 \\ -3 \\ 0 \\ 1 \end{pmatrix} s, \quad t, s \in \mathbb{R}$$

b) $\dim(N(A)) = 2$

c) Three vectors $a = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ -2 \\ -3 \\ 0 \\ 1 \end{bmatrix}$, $c = \begin{bmatrix} 0 \\ -2 \\ -2 \\ 1 \\ 1 \end{bmatrix}$

d) All inner products are 0 because the nullspace is orthogonal with the row space

Question 8 *(25 points)* Consider the following system of linear equation for some constant k:

$$x + y - z = -2$$
$$3x - 5y + 13z = 18$$
$$x - 2y + 5z = k$$

a) For which value(s) of k does this system have one or infinitely many solutions?

b) Find all solutions for each value of k.

a) 7.5 for row reduction
5 for finding correct k
b) 5 for $x_{particular}$
5 for $x_{nullspace}$
2.5 for x complete.

a) $\left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 3 & -5 & 13 & 18 \\ 1 & -2 & 5 & k \end{array}\right] \xrightarrow[R3=R3-R1]{R2=R2-3R1} \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & -8 & 16 & 24 \\ 0 & -3 & 6 & k+2 \end{array}\right] \xrightarrow[R3/3]{R2/-8}$

$\left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & 1 & -2 & -3 \\ 0 & -1 & 2 & (k+2)/3 \end{array}\right] \xrightarrow{R3=R3+R2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 0 & \frac{k+2}{3}-3 \end{array}\right]$

In order for the system to have infinitely many solutions

$\frac{k+2}{3} - 3 = 0 \Rightarrow k = 7$

So, for $k=7 \Rightarrow$ infinitely many solutions

b) - $x_{particular}$

Set $z = 0$

$z = 0$
$y = -3$
$x = 1$

$x_{particular} = \begin{bmatrix} 1 \\ -3 \\ 0 \end{bmatrix}$

- $x_{nullspace}$

$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ $z = 1$, $y = 2$, $x = -1$

$x_{nullspace} = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} s, \ \forall s \in \mathbb{R}$

$x = x_{particular} + x_{nullspace}$

$= \begin{bmatrix} 1 \\ -3 \\ 0 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} s, \ \forall s \in \mathbb{R}$

Question 9 *(5 points)* [Bonus] Please take a few minutes to share your thoughts on the course so far. Please be thoughtful and frank. Thank you.

- What do you like about the course so far?
- What suggestions do you have to improve this course?
- Do you have any other comments?